THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A student models a falling ball ignoring air resistance. For which object is the model least accurate? 1

(a) a steel bolt **(b)** an iron ball **(c)** a stone **(d)** a feather

2. A solution is prepared by dissolving 30 g of sugar in 120 g of water. The mass percentage is: 1

(a) 20 per cent **(b)** 12.5 per cent **(c)** 30 per cent **(d)** 25 per cent

3. In an experiment 5.6 g of CaO reacts with 4.4 g of CO2 to form CaCO3. The mass of CaCO3 formed is: 1

(a) 1.2 g **(b)** 9.6 g **(c)** 10 g **(d)** 11 g

4. An atom has the configuration 2, 8, 6. The number of protons is: 1

(a) 14 **(b)** 8 **(c)** 6 **(d)** 16

5. A car starting from rest has acceleration 3 m/s squared. Its velocity after 4 s is (m/s): 1

(a) 7 m/s **(b)** 12 m/s **(c)** 1 m/s **(d)** 6 m/s

6. A force of 24 N acts on a body and produces an acceleration of 4 m/s squared. The mass is: 1

(a) 6 kg **(b)** 0.17 kg **(c)** 20 kg **(d)** 96 kg

7. A force of 20 N acts at right angles to the displacement of 5 m. The work done is: 1

(a) 25 J **(b)** 4 J **(c)** zero **(d)** 100 J

8. The frequency of a wave is 500 Hz and the speed is 340 m/s. The wavelength is: 1

(a) 170000 m **(b)** 0.68 m **(c)** 1.47 m **(d)** 0.0015 m

9. A plant cell is placed in a solution and the cell contents shrink away from the wall. The solution is: 1

(a) hypotonic **(b)** pure water **(c)** isotonic **(d)** hypertonic

10. Which tissue would be most affected in a plant if the cambium is removed? 1

(a) gas exchange **(b)** water absorption
(c) increase in girth **(d)** increase in length

11. An organism is unicellular, eukaryotic and has chloroplasts and a flagellum. It belongs to: 1

(a) Fungi **(b)** Monera **(c)** Plantae **(d)** Protista

12. A farmer wants to grow a seedless banana plant. The method he uses is: 1

(a) grafting only **(b)** vegetative propagation by suckers
(c) cross-pollination **(d)** seed sowing

13. The distance from the Earth to the Sun is 1.5 x 10^11 m and the speed of light is 3 x 10^8 m/s. The time taken by sunlight is: 1

(a) 5 s **(b)** 500 s **(c)** 50 s **(d)** 5000 s

14. A pendulum model assumes no friction. The actual pendulum: 1

(a) speeds up **(b)** swings forever
(c) stops at once **(d)** slowly stops swinging

15. A mixture of two liquids X (b.p. 78 degrees Celsius) and Y (b.p. 118 degrees Celsius) is best separated by: 1

(a) crystallisation **(b)** sublimation
(c) filtration **(d)** fractional distillation

16. The formula of aluminium sulphate is: 1

(a) $AlSO_4$ **(b)** $Al(SO_4)_3$ **(c)** $Al_2(SO_4)_3$ **(d)** $Al_3(SO_4)_2$

17. An element has 3 shells and 2 valence electrons. Its atomic number is: 1

(a) 8 **(b)** 12 **(c)** 10 **(d)** 14

18. A body moves along a straight line with speed 4 m/s for 10 s and 6 m/s for the next 10 s. The average speed is: 1

(a) 10 m/s **(b)** 4 m/s **(c)** 6 m/s **(d)** 5 m/s

19. A person weighs 600 N on the Earth. On a planet with g one-third of the Earth's, the weight is: 1

(a) 600 N **(b)** 300 N **(c)** 1800 N **(d)** 200 N

20. A lamp of 60 W is used for 5 hours daily for 30 days. The energy consumed in kWh is: 1

(a) 9 **(b)** 1.8 **(c)** 90 **(d)** 300

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. A model assumes frictionless motion. Give one situation where it is acceptable and one where it fails. 2

22. Calculate the mass of sugar required to prepare 500 g of 8 per cent solution. 2

23. In an experiment 2.0 g of hydrogen reacts with 16 g of oxygen to form water. Find the mass of water and verify the law of conservation of mass. 2

24. An atom has 12 protons and 12 neutrons. Find its mass number and electronic configuration. 2

25. A car moves with uniform velocity of 15 m/s for 20 s. Find the distance and draw the nature of the v-t graph. 2

26. A body of mass 5 kg is acted on by two forces: 20 N east and 8 N west. Find the acceleration. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A student measures the time period of a pendulum by timing 20 oscillations as 32 s. Find the time period and say why 20 oscillations are timed. 3

28. A 200 g solution has 36 g salt. Find the percentage. If 100 g water is evaporated find the new percentage. 3

29. A compound of carbon and oxygen has C : O = 3 : 8 by mass. Find the amounts of C and O in 55 g of the compound. 3

30. An element has 20 protons and 20 neutrons. Find its mass number, electronic configuration, valency and name. 3

31. A bus moving at 20 m/s speeds up uniformly to 30 m/s in 5 s. Find the acceleration, the distance and the average velocity. 3

OR

A body of mass 10 kg is acted on by a force of 50 N for 4 s starting from rest. Find the acceleration, the final velocity and the distance covered. 3

32. A 2 kg body is thrown upward with 15 m/s. Find KE and PE at the start, at 5 m and at the top (g = 10). 3

OR

A source produces 4000 vibrations in 10 s. Find the frequency, the time period and the wavelength in air (v = 340 m/s). 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. A pendulum experiment gave: length (cm) 25, 50, 75, 100 and time for 10 oscillations (s) 10, 14.2, 17.4, 20. Calculate the time periods, comment on the relationship and mention the assumptions. 5

34. A student has three solutions A, B, C of the same solute with masses of solute 5 g in 95 g water, 10 g in 90 g water and 15 g in 85 g water. Find the percentage of each, rank them and say what happens on mixing A and C (equal masses). 5

35. Verify the law of constant proportions using water: 9 g of water gives 1 g hydrogen and 8 g oxygen, and 18 g of water gives 2 g hydrogen and 16 g oxygen. State the law. 5

36. An atom X has atomic number 13 and mass number 27. Find the electrons, protons, neutrons, configuration, valency, the ion it forms and the formula of its oxide. 5

OR

A car accelerates from rest at 2 m/s squared for 10 s, moves at constant velocity for 20 s and then retards uniformly to rest in 5 s. Find the maximum velocity, the retardation and the total distance. Draw the v-t graph in words. 5

37. A car of mass 1000 kg moves on a level road at 30 m/s. The engine force is 2000 N and the resistive force is 1500 N. Find the acceleration. If the engine is switched off and the resistance remains, find the retardation, the time to stop and the stopping distance. 5

OR

A body of mass 2 kg falls freely from 45 m. Find the PE and the KE at the start, after 1 s, 2 s and at the ground and show that the total energy remains constant (g = 10). 5

38. A person standing between two parallel cliffs claps. He hears the first echo after 1 s and the second after 2 s. Find the distance between the cliffs and the distances from the person (v = 340 m/s). What would be the time if he is at the centre? 5

OR

Describe the structure and functions of the nucleus, including the nuclear membrane, nucleolus, chromatin and nuclear pores. 5

END OF QUESTION PAPER **Total: 80 Marks**

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